Norms of communication in human–AI interaction: skin in the game and stake shape evaluations of human and artificial agents

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ABSTRACT

When information turns out to be wrong, does it matter whether the informant had something at stake, and whether they were a person or a machine? In a pre-registered between-subjects vignette experiment ($N = 355$; Prolific, UK), we tested how skin in the game and the stakes of an error shape evaluations of an assertion made by a human versus an artificial intelligence (AI) agent. An airport information-desk agent gave a passenger gate information that was correct when last recalled but had since silently changed. We crossed the severity of the consequence for the passenger (high vs. low) with the agent's identity (human with high skin in the game, human with low skin in the game, or AI) and measured the assertion's acceptability, justifiability, and willingness to rely on the agent again. Skin in the game increased reliance: the high-skin human was relied upon more than the low-skin human or the AI. The expected outcome effect appeared only in the consequence-sensitive judgments, forgiveness and the felt obligation to have warned, and responsibility shifted toward the passenger when the agent was an AI. We discuss what these asymmetries imply for norms of communication and for trust in human–AI interaction.

Introduction

People are increasingly receiving information from artificial systems, such as chatbots, voice assistants, and service robots, which answer questions, give directions, and make recommendations. When such a system provides information that later turns out to be wrong, how do people judge the mistake, and do they judge it in the same way as an equivalent mistake made by a person? These questions matter because reliance on an information source is not fixed: people revise it depending on who the source is and on the consequences of the error[1,2]. Understanding the terms on which people extend or withdraw reliance from human and artificial agents is central to designing interactions that align reliance to the agents' actual competence.

Recent work in human–AI interaction has emphasized giving users greater agency to understand, adapt, and shape AI systems, going beyond mere explanation or contestability[3]. We take a complementary, retrospective perspective by examining how people evaluate, rely on, and assign responsibility to human and artificial agents after an assertion, defined as a normatively governed speech act, turns out to be mistaken. In other words, when someone makes an assertion, they're taking on a responsibility for its truth, and can be held accountable if it turns out to be false.

A natural way to frame the evaluation of the provided information is through the norms that govern communication in the first place. On an influential account, communication is subject to a knowledge norm: one should assert only what one knows to be true[4]. In more permissible accounts, an agent who claims something false but who had good reason to believe it, for example, because it was true when last checked, does not violate communication norms, and observers may judge the agent as justified even when the content of the claim was incorrect. This distinction between the correctness of a claim and the epistemic standing of the agent who made it is the conceptual pivot of the present study, and it raises the question of whether observers apply the same standing-based leniency to a machine that they apply to a person[5].

Two further factors plausibly shape how the provision of the wrong information is evaluated. The first is what the agent personally stands to lose if the assertion is wrong, what Taleb[6] calls *skin in the game*. An agent who is personally exposed to the downside of an error has, according to this view, a credible incentive to get things right, and observers may reward that exposure with more lenient evaluations and greater willingness to rely on the agent again. The second is the severity of the consequences that actually follow from the error. There has been quite some research on the concept of *outcome effect* which indicates that people judge the quality of a decision partly by how it turned out, holding the information available to the decision-maker constant[7]. If observers exhibit an outcome effect here, an identical mistaken claim should be judged more

harshly when its consequences for the recipient are severe than when they are mild.

Whether these factors operate in the same way for human and artificial agents is, however, unclear. Skin in the game is defined in terms of a first-person stake, such as a livelihood, a reputation, or a future, none of which has a clean analogue for a service robot. A growing body of work also shows that people treat machine agents differently from human agents when it comes to error and responsibility: they can be averse to relying on algorithms after seeing them make an error[8,9], yet under other conditions prefer algorithmic to human judgment[10]; and they are reluctant to accept machines as moral decision-makers[11,12] and attribute responsibility to them along different lines than to people[13,14]. It is therefore an open empirical question whether and to what extent the leniency associated with an agent's stake in the outcome and epistemic standing emerges in this context, whether it extends from human to artificial agents, and whether outcome effects differ across agent types. (As explained in the Methods, we do not apply the notion of "skin in the game" to the AI agent; the AI condition serves as a reference level in which skin in the game is absent by construction.)

The present study addresses this gap using a pre-registered between-subjects vignette experiment. We ask: *How do skin in the game and the stakes of an error affect the evaluation of a claim made by a human versus an AI agent?* An information-desk agent gives a passenger gate information that was correct when last checked but has since silently changed. We cross the stake of the error for the recipient (high vs. low) with the identity of the agent (a human with high skin in the game, a human with low skin in the game, or an AI) and measure how acceptable the claim was perceived, how justified the agent was in making it, and how willing people would be to rely on the agent again. Because employment-based skin in the game has no clear analogue for the AI, we treat the AI as a clean reference level rather than forcing an artificial skin manipulation (see Methods).

Prior literature on norms of communication allows us to formulate the following hypotheses. **H1:** Human agents with high skin in the game receive more lenient evaluations, including higher acceptability, justifiability, and reliance, than human agents with low skin in the game. This prediction is motivated conceptually by Taleb[6] and by evidence that costly, credible signals of commitment raise perceived trustworthiness[15,16]. To our knowledge, however, no prior study manipulates an agent's own downside exposure and measures how observers evaluate that agent's error; establishing whether exposure buys leniency, and on which dimensions, is part of this study's contribution. **H2:** The AI is judged similarly to a human with low skin in the game[8,10,11,17]. **H3:** Errors with severe consequences (high stakes) are judged less acceptable and less justified than errors with mild consequences (an outcome effect)[7,18–20]. **H4:** This outcome effect is attenuated for human agents with high skin in the game[21–23]. As we report below, the data support H1 for reliance and broadly support H2, but the predicted outcome effect (H3) does not appear for human agents on acceptability or justifiability and emerges only in the consequence-sensitive judgments (forgiveness and the felt obligation to have warned), while several unanticipated asymmetries between the human and AI agents came to light: although a competent AI agent is evaluated similarly to a low-skin human agent in some respects, it is relied upon and held responsible differently.

These findings contribute to our understanding of human–computer interaction by showing that the norms governing reliance on and responsibility attribution toward information sources do not transfer straightforwardly from human to artificial agents. In particular, competence alone does not guarantee reliance on an AI system, as reliance also depends on users' beliefs about the system's access to current information. Moreover, responsibility for AI errors tends to shift away from the system and toward the user who acts on its advice. These asymmetries have concrete implications for the design of AI information services, which we develop further in the Discussion: an accurate-on-average system cannot assume that it will be relied upon, and interfaces may need to make an agent's information provenance and update status explicit rather than leaving users to infer them.

Background

Norms of communication

As previously mentioned, an assertion is a normatively governed speech act: in telling someone that something is true, an agent represents themself as being in a position to do so. In the knowledge account, the norm is that one should assert only what one knows[4]. Because knowledge is factive, as one cannot know something that is factually false, this account entails that a properly warranted assertion must also be true. Justification and truth cannot come apart. This factivity requirement has, however, been empirically challenged. Knowledge is a poor predictor of whether an assertion is judged appropriate. On this account, justification, and truth do not need to coincide. An agent may reasonably claim information that later turns out to be false or outdated because circumstances have changed, without having violated the norm of assertion at the time of speaking[5]. As conversational AI systems increasingly provide factual information, users evaluate their statements in ways that resemble the evaluation of human statements. Our study is built around exactly this case: the agent provides information upon inquiry that was correct at the time of the last recall but has been changed by an undocumented update. This design lets us ask whether people distinguish the correctness of an assertion from the standing of the asserter and, critically, whether they grant that distinction to an artificial agent as readily as to a human one. Whether a machine literally performs an assertion, rather than emitting a signal that observers interpret as one, is up for debate; we remain agnostic on this question and treat the

AI's utterance as a functional assertion that participants evaluate as they would a human's. A growing experimental literature operationalizes norms of assertion by eliciting lay judgments of whether an agent's claim was proper or warranted given what they knew[5,24]. Turri[24] finds that such judgments track an assertion's truth even when the speaker had good evidence for it, supporting a factive norm. Kneer's later data[5] challenge this, showing knowledge to be a poor predictor of assertability and favoring a justified-belief account instead. We adopt the *justifiability* item from this paradigm, and extend it, alongside an acceptability item, to a direct comparison of human and AI agents, capturing whether the agent had good reason for the claim, independent of whether it was true[25].

Skin in the game

Taleb[6] argues that agents who bear personal consequences for their actions behave, and may be evaluated, differently from agents who do not: personal exposure to loss disciplines judgment and signals trustworthiness. Translated into the evaluation of an assertion, the idea predicts that an agent who has something to lose if their claim proves wrong (a job, a reputation, a future opportunity) should be extended more latitude for an honest mistake and should attract greater willingness to be relied upon again. The idea of personal consequences from economics resonates with the commitment to assertion via the *liability account*, well explored conceptually in philosophy (see, e.g., ref.[26]). On this view, making an assertion is a bit like offering a warranty: the speaker takes on responsibility for what happens if the hearer relies on the claim and it turns out wrong. This is a separate question from whether the speaker met the knowledge or justification norm discussed above. A speaker can be justified in what they claimed and still be held liable if someone is harmed by relying on it. Skin in the game captures exactly this liability side: it tells us how much the speaker personally has to lose, separately from whether their claim was epistemically sound. Our key contribution to this literature is the empirical test of relevance of personal consequence for the agent for the evaluation of the AI and human assertions. Skin in the game is inherently a first-person notion: it refers to what *the agent* stands to lose. This is what makes its extension to artificial agents problematic. Can an autonomous system be held liable at all? Some argue that responsibility for a machine's error gets spread across so many parties (designers, deployers, operators) that no one is left clearly answerable for it, a "responsibility gap". Relatedly, when an AI is involved, blame tends to shift onto the system itself rather than staying with the people behind it[27]. If it's unclear who, or what, can even be held liable when an AI is involved, then skin in the game may not make conceptual sense for AI at all. A service robot does not have a livelihood or a career to forfeit, and manufactured substitutes, for example, being "switched off" or a reputational cost borne by its operator, do not reproduce the agent's own exposure. Threatening to switch a machine off doesn't recreate what Taleb's account is really about, namely the personal fear of losing something that disciplines a human's judgment. And pinning the cost on the operator's reputation just moves the stake to a different party altogether, not the one who actually made the claim. We come back to this gap in the Discussion, where we suggest ways future work could test artificial stand-ins for stake (like penalties tied to service agreements, or public reliability scores) to see whether it's exposure itself, or simply being human, that earns an agent more leniency. We therefore manipulate skin in the game only within the human conditions and use the AI as a reference level, treating the comparison of a low-skin human with an AI as a test of how far the construct's consequences travel beyond the human case.

Outcome effect and the evaluation of decisions

People routinely let the consequences of a decision color their evaluation of the decision itself. In the classic demonstration, participants judged decisions that led to bad outcomes more harshly than identical decisions that happened to turn out well, even when the information available beforehand was the same[7], a pattern originally termed *outcome bias*. Outcome information continues to shape moral judgment even when intentions are held fixed[18–20]. This outcome effect motivates our stake manipulation: we hold the agent's information and action constant and vary only the severity of the downstream consequence for the recipient. If observers evaluate the assertion partly by its consequences, the same mistaken assertion should be judged less acceptable and less justifiable when the recipient suffers a serious loss than when the loss is averted. Our design therefore allows us to examine if this outcome effect interacts with agent type and perceived personal stakes or if it is moderated by whether the agent is a person, who might be excused a costly error in light of their exposure, or a machine.

Trust and reliance in human–AI interaction

Reliance on automation is well understood to depend on trust that is continually recalibrated in response to the system's performance, and mismatches between trust and actual reliability lead to misuse or disuse[1]. Integrative and meta-analytic reviews further show that trust in automated and robotic agents is shaped jointly by characteristics of the human, the system, and the context[28,29]. How people revise reliance after an error, however, differs between human and algorithmic agents in ways that resist a single summary. On the one hand, people can be disproportionately quick to abandon an algorithm once they have seen it make a mistake, even when it outperforms a human alternative[8]. On the other hand, when forming judgments, people sometimes weight algorithmic advice more heavily than equivalent human advice[10]. Reviews of the empirical literature emphasize that trust in AI is shaped by the agent's perceived characteristics, including its transparency and its apparent access

to relevant information, and not by accuracy alone[2], and by whether and how the source of the advice is disclosed[30]. These findings lead us to expect that a mistaken assertion will not move reliance in the same way for an AI as for a human. Trust may be shaped not only by the presence of an error but also by whether the source is perceived to share in the risks and consequences associated with the decision. By focusing on perceived exposure to decision consequences, our study extends the trust literature beyond traditional concerns with competence, reliability, and benevolence, highlighting "skin in the game" as a distinct factor shaping reliance on AI versus human advisors.

A recurring theme in this literature, sharpened by recent work in human–computer interaction and explainable AI, is that *trust* (an attitude) and *reliance* (a behavior) are distinct constructs that need not move together and are often conflated or measured with instruments of uncertain validity[31,32]. Appropriate reliance, on this view, is a matter of *calibration*: aligning reliance with the agent's actual competence, which itself shifts as the agent's performance and the confidence signals it emits change over the course of interaction[32,33]. Our design captures a single post-error snapshot rather than this longitudinal dynamic; we measure reliance (DV3) as a self-reported behavioral intention and treat acceptability and justifiability as attitudinal judgments about the assertion.

In empirical human–computer interaction and human-factors research, reliance is accordingly often operationalized behaviorally rather than by self-report: participants complete incentivized decision tasks with an AI recommendation available, and reliance is read off whether they accept, override, or verify that recommendation, with trust measured separately by validated multi-item scales[34–37]. A complementary, decision-theoretic strand formalizes the same target as a *rational benchmark*, the reliance policy as a payoff-maximizing agent would adopt given the AI's and the person's respective accuracies, and scores observed behavior as over- or under-reliance relative to that optimum rather than against a fixed threshold[38]. A systematic overview of this empirical literature documents how widely such objective, outcome-referenced reliance measures have displaced self-report in AI-assisted decision-making studies[39], and controlled experiments now estimate both the extent of over-reliance and its performance cost directly from choice behavior[40]. Our single-item reliance intention captures none of this case-level structure: it registers how willing a participant says they would be to consult the agent again, not whether that willingness would be calibrated to the agent's competence on the next question, a gap we take up in the Discussion. Trust instruments, by contrast, are multidimensional by design. The widely used scale by Jian et al.[34] spans twelve items covering reliability, predictability, and confidence, and Muir and Moray[35] track how trust develops with observed competence over repeated interactions. Scale development has continued since: the Trust in Automation questionnaire of Körber[41] grounds six facets (including competence, predictability, and propensity to trust) in interpersonal-trust theory, and the Trust of Automated Systems Test offers a brief two-factor (understanding, performance) instrument with initial psychometric validation[42]; either would be a drop-in replacement for our single items in a follow-up study. A methodological survey of empirical human–AI trust studies nonetheless found that many measure trust with unvalidated ad hoc items and frequently conflate trust with reliance and recommends stating explicitly which construct each item is meant to tap[43]. We follow that recommendation for a design whose measures predate it: our reliance item (DV3) is a behavioral intention, our acceptability and justifiability items (DV1–DV2) are normative judgments about a specific assertion rather than facets of a trust scale, and none of the three is a validated multi-item trust instrument, a limit stated where the measures are introduced (Methods) and weighed in the Discussion. This paradigm has produced two findings that frame our design. First, people frequently over-rely, accepting incorrect recommendations they could have caught, and interface interventions that prompt verification before acceptance, such as cognitive forcing functions, measurably reduce this over-reliance at some cost in speed and satisfaction[44]; this risk is not confined to laboratory tasks, as explanations intended to support trust calibration in deployed clinical decision-support systems have themselves been observed to introduce over-reliance on the AI's recommendation[45], and incorrect AI advice has been shown to degrade the quality of human decisions even when the advice is accompanied by explanations[46]. The errors of an AI can moreover outlive the interaction itself: people who performed a medical-style classification task with a systematically biased AI went on to reproduce its errors in their own subsequent unassisted judgments[47]. Whether people verify at all follows a cost-benefit logic, with over-reliance dropping when the effort cost of checking is lowered or the incentives for accuracy are raised[48]. Second, whether people verify at all is itself a behavior of interest: the division of checking labor between human and AI is increasingly treated as a cooperative-verification problem, supported by provenance cues such as reference links and update indicators that let users audit where an answer came from[49,50]. The need for such verification support is not hypothetical: a systematic audit of commercial generative search engines found that only about half of generated sentences were fully supported by their own citations[51], so even the provenance cues users are shown can be unreliable, making the recipient's independent check, the behavior embedded in our vignette, a live component of real-world reliance. The same paradigm underpins the trust-calibration literature on over- and under-reliance, where what is manipulated is typically the AI's displayed confidence and the form of its explanations, and what is measured is acceptance behavior and complementary human–AI team performance[36,37]; recent extensions to conversational agents show that people systematically overestimate the reliability of large-language-model answers relative to the model's own internal confidence, a calibration gap that widens or narrows with how the model's confidence is verbalized[52], and that first-person expressions of uncertainty by such a model

measurably reduce users' over-reliance[53]. Our vignette places the verification behavior inside the stimulus, since the recipient either double-checks the gate board or does not, and its assertion is an unhedged, maximally confident answer from an agent whose actual epistemic position did not warrant that confidence; our reliance item, by contrast, elicits a single self-reported intention after a described error. We therefore read our reliance measure as a snapshot proxy for these behavioral paradigms, not a substitute for them, and we return to this limit in the Discussion. Finally, reliance and trust are moderated by stable individual differences. Dispositional trust in automation shapes reactions to machines most strongly before and early in an interaction, precisely the single-encounter situation our vignette describes, and moderates how people respond to observed errors[29,54]; familiarity with AI systems and AI literacy, the cluster of competencies needed to evaluate and critically appraise AI outputs, likewise moderate reliance[55], and validated instruments now exist for the algorithmic-literacy component, covering awareness of algorithmic curation and knowledge of how algorithmic systems work[56]. Our design randomizes these across conditions but does not measure them, a point we take up in the Discussion.

Responsibility attribution to human and artificial agents

Finally, mistaken assertions invite judgments of responsibility and blame, and these judgments follow structured rules. They depend on perceived causation, intentionality, and the reasons available to the agent[13]. When the agent is artificial, these attributions shift. People perceive the minds of machines more readily along a dimension of *agency*, the capacity to plan and act, than along a dimension of *experience*, the capacity to feel[57], and they anthropomorphize systems more strongly when those systems are framed as autonomous, thinking agents[58]; subsequent work suggests lay conceptions of mental life are richer still, distinguishing suites of capacities related to the body, the heart, and the mind[59]. Framing effects of this kind are experimentally established. Describing an autonomous machine in anthropomorphic, mind-having terms increases trust in it and changes how its errors are attributed[60], and when an AI mediates an interaction, responsibility can be displaced onto the intervening system, which absorbs blame like a "moral crumple zone"[27]. Because responsibility attribution tracks perceived agency, how an AI is described, for example, as capable of independent decisions, can itself move how much responsibility observers assign to it, a consideration that motivates our attention to the agent's framing (Methods). People are reluctant to accept machines as moral decision-makers[11] and hold distinctive expectations about how machines should resolve morally loaded trade-offs[12]. They may also distribute responsibility differently between a machine agent and the surrounding humans, a redistribution that connects to debates over a "responsibility gap" for autonomous systems, where no party cleanly bears responsibility for a machine's error[61]. Experimental work on reactions to algorithmic harm confirms that observers do divide blame between machines and the people around them, with the division tracking how the machine and its role are perceived[14]; our design measures exactly this division, between the asserting agent and the recipient who acted on the assertion. Computational accounts make the structure of such judgments explicit: formal definitions of blameworthiness locate it in an agent's causal contribution to an outcome given what that agent was in a position to know[62], and people's causal judgments about physical events are well captured by counterfactual simulation, asking whether the outcome would have differed had the cause been absent[63]; we extend this framing to an agent's assertion. Viewed through this lens, our stake manipulation changes the recipient's counterfactual contribution, since in the high-stake setting, the recipient could have double-checked the information and did not. This both explains the uniformly high level of attributed responsibility assigned to the recipient and makes explicit why outcome severity and verification behavior cannot be disentangled in this design (Methods). We therefore measure responsibility for the outcome separately for the agent and the recipient, allowing us to detect whether blame shifts toward the recipient when the agent is an AI. Beyond assigning responsibility, people may also differ in their willingness to forgive an agent following an error. We also measure *forgiveness*, the willingness to look past a mistake and continue interacting[64], as a complementary route through which skin in the game and outcome severity might operate.

Results

Of the 416 participants who completed the survey, 61 were excluded for failing the first attention check (the pre-registered exclusion rule), leaving a final analytic sample of $N = 355$ that was approximately balanced across the six conditions (per-cell $n = 55$–63; 177 high-stake and 178 low-stake; see Table 1). We first report the manipulation checks, then the confirmatory analyses of the three primary outcomes, followed by the exploratory analyses of the secondary outcomes, the database-belief grouping, and the qualitative analysis of the open-ended responses. All statistics are reported as computed in the pre-registered analysis plan; effect sizes are Cohen's d for pairwise contrasts and partial η^2 for omnibus effects.

Four design choices, introduced and justified in the Methods, carry direct implications for how specific results should be read, and rather than deferring them to the limitations we flag each at the point where it bites: the acceptability item's wording and floor where the acceptability results are reported, the coupling of outcome severity with the recipient's verification behavior where the stake effect on forgiveness is reported, the responsibility items' outcome-anchored phrasing and the AI vignette's autonomy framing where the responsibility attributions are reported, and the vignette's "usually departs" cue where the diligence perceptions are reported. Supplementary analyses that bound each concern accompany the relevant result.

Manipulation checks

Both manipulations were highly successful, with very large effect sizes (all Cohen's $d > 1.0$; Supplementary Table S1). Within the human conditions, participants in the high-skin condition rated Ken as caring more about the supervisor's assessment, as more likely to feel bad about providing wrong information, and as having more to lose than participants in the low-skin condition. The stake manipulation was likewise successful: participants in the high-stake condition perceived the negative effect on Ms. Miller as substantially stronger than participants in the low-stake condition. Participants thus clearly distinguished both the level of skin in the game in the human conditions and the severity of the consequences for Ms. Miller.

Primary dependent variables

The primary outcomes were acceptability, justifiability, and reliance. Omnibus 2 (Stake) $\times$ 3 (Agent) ANOVA results are summarized in Table 2, cell and condition means in Supplementary Table S2, and the condition-by-stake patterns are shown in Fig. 1.

Acceptability (DV1). The analysis showed a significant main effect of stake and a significant Stake $\times$ Agent interaction, but no significant main effect of agent. The interaction was driven entirely by the AI condition: participants rated Ken's answer as more acceptable in the AI high-stakes condition than in the AI low-stakes condition ($M = 3.00$ vs. $M = 1.89$, $p < .001$, $d = 0.63$). The human high-skin and human low-skin conditions showed essentially no difference between the high- and low-stake scenarios. This pattern is the opposite of the predicted outcome effect. Acceptability also showed a strong floor effect, with roughly 47% of participants selecting the lowest scale point (Supplementary Fig. S2). Two measurement features therefore qualify all acceptability results as they are reported: the floor compresses variance and limits sensitivity to condition differences, and the item's wording (whether Ken *should have said* the outdated gate number) may have cued the factual incorrectness of the answer rather than the epistemic standing of the agent who gave it. Both considerations, developed in the Discussion, favor justifiability as the cleaner index of the assertion norm. A supplementary floor-composition check supports reading the floor as a property of the item rather than of any condition: in logistic models of the probability of answering at the lowest scale point, neither stake, nor agent, nor their interaction was significant (likelihood-ratio tests, all $p \geq .30$), and the share of at-floor responses is similar in every cell (Supplementary Fig. S1), so the condition effects on mean acceptability reported above are carried by the responses above the floor.

Justifiability (DV2). There was a significant main effect of agent and a significant Stake $\times$ Agent interaction, but no significant main effect of stake. Pairwise comparisons indicated that the human high-skin condition was rated as more justified than the AI condition (Bonferroni $p = .010$, $d = 0.38$). As with acceptability, the interaction was driven by the AI condition, where justifiability was higher under high than low stake ($M = 3.97$ vs. $M = 3.31$, $p = .035$); the human conditions showed the opposite, non-significant descriptive trend. The omnibus interaction test was part of the a priori plan; the within-condition simple effects that decompose it were not, and are treated as exploratory (see 'Exploratory findings').

Reliance (DV3). There was a significant main effect of agent, but no significant main effect of stake and no significant interaction. Collapsed across stake, participants were most willing to rely on Ken in the human high-skin condition ($M = 3.23$), followed by the human low-skin condition ($M = 2.66$) and the AI condition ($M = 2.50$). Bonferroni-corrected contrasts (Supplementary Table S3) showed that reliance was significantly higher in the human high-skin condition than in both the human low-skin condition ($p = .021$, $d = 0.36$, 95% CI $[0.10, 0.62]$) and the AI condition ($p = .001$, $d = 0.46$, 95% CI $[0.20, 0.71]$); the human low-skin and AI conditions did not differ ($p = 1.000$, $d = 0.10$, 95% CI $[-0.16, 0.36]$, an interval that includes zero). A supplementary equivalence analysis (two one-sided tests; not part of the a priori plan) supported the equivalence of the low-skin human and AI conditions on reliance within a smallest-effect-of-interest bound of Cohen's $d = 0.4$, providing positive evidence that these two conditions were treated similarly rather than merely a non-significant difference. Extending the same analysis across the outcomes sharpens where the human–machine parallel holds and where it breaks: the two conditions were likewise statistically equivalent on acceptability, equivalence could not be established for justifiability (observed $d = 0.24$, TOST $p = .10$, an indeterminate result: neither reliably different nor demonstrably equivalent), and it was clearly ruled out for forgiveness (observed $d = 0.41$), where the AI is forgiven significantly less (see 'Secondary dependent variables'). The reliance equivalence does not hinge narrowly on the choice of bound: the observed contrast is small ($d = 0.10$) and its 95% confidence interval $[-0.16, 0.36]$ lies within the ± 0.4 region, so a moderately stricter bound would preserve the conclusion, whereas the indeterminate justifiability contrast ($d = 0.24$) would remain indeterminate; the supplementary script reports the tightest bound at which each equivalence still holds. A Bayesian counterpart to these tests (JZS default prior, $r = 0.707$), which frames the low-skin-human vs. AI comparison in the same terms as the outcome-effect nulls reported under 'Exploratory findings', returns the same four verdicts: moderate evidence for equivalence on reliance ($BF_{01} = 5.3$) and acceptability ($BF_{01} = 3.8$), an inconclusive result for justifiability ($BF_{01} = 1.5$), and clear evidence for a difference on forgiveness ($BF_{10} = 15$). Among the primary outcomes, reliance therefore provided the clearest support for the skin-in-the-game prediction.

Hypothesis evaluation. Table 3 summarises the outcome of the four a priori hypotheses. In brief, skin in the game increased reliance (H1, partially supported: HHS > HLS on reliance only); AI was judged like a low-skin human on acceptability and reliance (H2, largely supported); the predicted outcome effect did not appear for humans and reversed for AI (H3, not supported for the primary outcomes); and H4 could not be evaluated because no outcome effect was present for humans on acceptability or justifiability.

Secondary dependent variables

Warning obligation (DV4). Participants strongly endorsed the statement that Ken should have pointed out the possibility of a gate change, in every condition (cell means 5.33–6.11 on the seven-point scale; Supplementary Fig. S3). The 2 (Stake) $\times$ 3 (Agent) ANOVA showed a significant main effect of stake, $F(1,349) = 11.61$, $p < .001$, $\eta^2 = .032$: participants in the high-stake condition endorsed the warning obligation more strongly than those in the low-stake condition ($M = 6.05$, $SD = 0.99$ vs. $M = 5.63$, $SD = 1.31$; $d = 0.35$). Neither the agent effect ($F = 2.10$, $p = .124$) nor the interaction ($F = 1.20$, $p = .301$) was significant. Two aspects of this result are notable. First, the felt obligation to hedge an assertion applied with equal force to the human and AI agents; participants did not hold the machine to a laxer disclosure standard. Second, outcome severity *retroactively* strengthened the perceived obligation: the same unhedged assertion was judged more deficient in hindsight when its consequences turned out to be severe. This places the warning obligation alongside forgiveness as the second consequence-sensitive judgment in the study, a pattern we return to in the Discussion.

Forgiveness (DV5). Forgiveness showed the strongest and clearest effects among all outcomes (Supplementary Table S4; Fig. 2). Both stake and agent had significant main effects, and the interaction was not significant. Mistakes in the low-stake conditions were forgiven far more than in the high-stake conditions ($M = 4.90$ vs. $M = 3.67$), and forgiveness followed a clear condition gradient, HHS (4.94) > HLS (4.40) > AI (3.55). The stake effect was significant within every condition: HHS ($d = 0.85$), HLS ($d = 0.68$), and AI ($d = 0.36$). Compared with the primary outcomes, forgiveness showed by far the clearest effect of outcome severity. One design feature bears on how this stake effect should be read: the stake manipulation varied the recipient's own verification behavior together with the outcome (in the low-stake vignette, Ms. Miller double-checks the gate board; in the high-stake vignette, she does not; see Methods). The forgiveness difference between stake conditions therefore reflects the joint effect of a worse outcome and an unverified reliance, and we cannot apportion it between the two; we flag this confound wherever stake effects on forgiveness and responsibility are interpreted (Discussion). A partial decoupling is nonetheless possible within the data: inside each stake condition, the recipient's verification behavior is constant (every low-stake recipient double-checked; no high-stake recipient did), while the perceived severity of the consequence varies across participants. Forgiveness tracked perceived severity within both conditions ($r = -.21$, $p = .006$ in the high-stake cells and $r = -.21$, $p = .005$ in the low-stake cells), indicating a severity component that operates independently of the recipient's verification behavior; how much of the between-condition difference this component explains, however, cannot be settled by this design.

Responsibility attribution (DV6). Ken's attributed responsibility differed significantly by agent condition, $F = 7.39$, $p < .001$, $\eta^2 = .041$: participants attributed the most responsibility to Ken in the human high-skin condition ($M = 5.52$) and the least in the AI condition ($M = 4.65$), with the human low-skin condition intermediate ($M = 5.23$). By contrast, Ms. Miller's attributed responsibility showed no significant condition effect, remaining between about 5.5 and 5.7 across conditions. The responsibility gap (Ken minus Miller) also differed by condition, $F = 4.48$, $p = .012$: it was effectively zero for the high-skin human (+0.05, n.s.), negative for the low-skin human (-0.50, $p = .040$), and most negative for AI (-0.92, $p < .001$). When Ken was an AI, blame shifted significantly toward the passenger (Fig. 3).

Two features of the design bear directly on these attributions. First, the responsibility items asked how responsible each party was for Ms. Miller "arriving at the correct gate on time," a phrasing that sits awkwardly in the high-stakes cells, where she did not arrive on time, and may have added interpretive noise there; the agent effects reported above collapse across stakes, which dilutes but does not eliminate this concern (see Discussion). A supplementary check confines the concern further: within the low-stake cells alone, where Ms. Miller did arrive on time and the phrasing is unproblematic, the recipient-ward shift for the AI remains clear (gap $= -1.11$, $t(60) = -3.27$, $p = .002$; Bonferroni-corrected across the three conditions, $p = .005$) while the high-skin human's gap remains near zero (-0.30, n.s.). The item wording therefore cannot account for the agent asymmetry, although the halved per-cell samples make these within-stake tests correspondingly less powerful. Second, the AI vignette explicitly described the agent as "fully capable of taking independent decisions" (Methods), a framing that, on mind-perception accounts, should if anything *inflate* the agency attributed to the AI and hence the responsibility assigned to it[57,58]. That participants nevertheless assigned the AI the *least* responsibility means the observed shift of blame toward the recipient is unlikely to be an artifact of an under-agentive framing; the framing biases against the effect we observe, and a more tool-like description would plausibly have shifted responsibility toward the recipient even further.

Two further exploratory observations bear on these attributions. Paralleling the forgiveness analysis, perceived severity predicted Ken's responsibility within the high-stake cells, where verification behavior is constant ($r = .17$, $p = .025$; low-stake

cells $r = .06$, n.s.), again indicating a severity component that operates independently of verification. And within the AI condition, individual differences in how connected and up-to-date the machine was perceived to be were unrelated to the responsibility assigned to it ($r = .04$, n.s.), suggesting that the AI's reduced responsibility is categorical, attached to the kind of agent it is, rather than scaling with its perceived capability; a graded agency-construal account would have predicted a positive association.

Database-connectivity perceptions (DV7). The database-connectivity items revealed what we term an *AI perception paradox* (Supplementary Table S5; Fig. 4). Participants rated the AI as more likely than the human agents to have checked the database ($d = 0.43$), yet much lower on the immediate availability of updated information ($d = -1.15$, the largest effect in the study) and lower on awareness of updates ($d = -0.39$). Ratings of direct connection to the database did not differ significantly between AI and human agents ($d = 0.14$). Participants thus did not simply perceive the AI as less diligent, but rather as less connected to up-to-date information: the items appear to tap two distinct facets, diligence (whether the agent checked the database) and live connectivity (whether updates are available to and known by the agent). These are empirically separable facets rather than one blurred construct: perceived diligence was essentially uncorrelated with the connectivity items ($r = .10$ with their mean), which themselves cohered as a subscale (Cronbach's $\alpha = .61$), and a components analysis of the four items returned two eigenvalues above one (1.72 and 1.20); future work could nonetheless sharpen the boundary with dedicated multi-item scales. The low absolute diligence ratings across all agents (2.01–2.79 on a seven-point scale) should also be read against the vignette's wording: the agent answers on the basis of where the flight "usually" departs (Methods), a phrase that participants may have taken as signalling habit rather than a recent check, depressing perceived diligence for human and AI agents alike; the *relative* agent differences reported here are unaffected by this common shift.

Taken together, the secondary outcomes indicate that outcome severity was reflected in the two consequence-sensitive judgments, forgiveness and the warning obligation, whereas responsibility attribution and database-connectivity perceptions differed mainly by agent condition (Figs 3 and 4).

Correlational structure

Across the full sample, believing that Ken had checked the database correlated positively with acceptability ($r = .44$, $p < .001$); justified mistakes were forgiven more ($r = .45$); and forgiveness predicted willingness to rely on Ken again ($r = .39$). Acceptability and justifiability were related but distinct ($r = .33$), as were justifiability and reliance ($r = .29$). Expecting a warning was associated with lower acceptability ($r = -.16$, $p = .003$), and there was a slight blame trade-off between the two responsibility judgments ($r = -.16$, $p = .003$). The forgiveness–reliance association was markedly stronger for low-skin humans ($r = .57$) than for high-skin humans ($r = .29$) or AI ($r = .22$): when an agent has little to lose, whether one forgives a mistake almost entirely determines whether one would rely on them again. Perceived diligence (checking the database) strongly predicted acceptability for AI ($r = .50$) and low-skin humans ($r = .54$), but less so for high-skin humans ($r = .25$).

The full inter-item correlation matrix, computed on the deposited dataset, is shown in Supplementary Fig. S4 and is consistent with the measurement model set out in the Methods: the epistemic-conduct judgments correlate moderately with each other and with reliance, the attribution and mental-model items form their own weakly connected blocks, and no pair of items approaches the redundancy that would suggest a single evaluative factor. The two consequence-sensitive judgments, forgiveness and the warning obligation, are negatively related at the person level ($r = -.15$): both respond to outcome severity, but in opposite directions, severe outcomes reducing forgiveness while increasing the demanded precaution. Because the matrix is computed on the deposited dataset with pairwise-complete observations, two coefficients differ by up to .02 from the values quoted above, which are reported as computed in the pre-registered analysis plan.

Exploratory findings

An AI-specific high-stake pattern (exploratory). Within the AI condition, high-stake participants rated the AI as more connected to the database ($M = 4.03$ vs. 3.16, $p = .007$), more likely to have checked it (3.11 vs. 2.46, $p = .044$), and its belief as more justified (3.97 vs. 3.31, $p = .035$), yet simultaneously as less forgivable (3.21 vs. 3.90, $p = .041$). When consequences are severe, participants may engage in compensatory reasoning, attributing greater diligence to the AI while still holding it accountable.

Composite leniency and robustness. A standardized composite of acceptability, justifiability, reliance, and forgiveness (Cronbach's $\alpha = .64$) confirmed the overall gradient, HHS ($M = 0.18$) > HLS (-0.02) > AI (-0.16), $F = 7.65$, $p < .001$. Non-parametric Kruskal–Wallis tests confirmed the parametric agent effects for justifiability ($H = 9.36$, $p = .009$), reliance ($H = 14.43$, $p < .001$), and forgiveness ($H = 29.02$, $p < .001$); the agent effect on acceptability was not significant non-parametrically ($H = 0.86$, $p = .652$), consistent with its floor effect. As a further robustness check, the confirmatory 2×3 ANOVAs for the three primary outcomes were re-estimated with participant age and gender entered as covariates; the condition and stake effects were unchanged in direction and statistical significance, indicating that the primary findings are not attributable to demographic composition.

The exploratory analyses in this section (the database-belief grouping, the within-AI high-stake contrasts, the composite index, and the qualitative cross-tabulations) were not part of the a priori confirmatory plan and were not corrected for multiple comparisons; the individual p-values should therefore be read as descriptive and hypothesis-generating rather than as confirmatory tests. As a formal sensitivity analysis, we applied a Holm–Bonferroni correction across the family of eighteen exploratory tests reported in this paper (computed on the deposited dataset). Every exploratory effect on which our interpretation rests survives, all at adjusted $p < .01$: the stake and agent effects on forgiveness, the stake effect on the warning obligation, the agent effect on Ken's responsibility, the AI condition's agent–recipient responsibility gap, the agent effect on the composite index, and the AI-vs-human perception differences on the checked-database, availability-of-updates, and awareness items. By contrast, the within-AI high-stake contrasts ($p = .007$–$.044$ uncorrected), the omnibus agent effect on the responsibility-gap difference score ($p = .012$), and the responsibility-gap decomposition for the low-skin human ($p = .040$) do not survive correction and should be treated strictly as hypothesis-generating. This caution applies with particular force to the within-AI high-stakes contrasts (the "AI high-stakes paradox"), which are based on cell sizes of roughly 30 participants and several p-values close to the .05 threshold.

We ran four further robustness checks on the primary outcomes. First, to guard against the ordinal, floored nature of the response scales, we re-estimated the agent effects with proportional-odds ordinal logistic models; these reproduced the parametric pattern, with the agent effect non-significant for acceptability ($\chi^2(2) = 0.98$, $p = .61$, consistent with its floor) and significant for justifiability ($\chi^2(2) = 9.60$, $p = .008$) and reliance ($\chi^2(2) = 14.61$, $p < .001$). Second, all participants in the analytic sample answered the agent-identity comprehension item correctly, so the primary results do not depend on comprehension-based exclusions. Third, a formal mediation of the agent effect on reliance through the database-connectivity perceptions was inconclusive: because those perceptions are strongly collinear with agent type (functioning in part as checks on how the agent was construed), the mediation model was unstable, so we report the perception differences descriptively (see 'Secondary dependent variables') rather than as a mediating mechanism. Fourth, we tested whether the agent effects were moderated by the two demographic variables available (each participant's demographic record was rejoined from the raw survey export, with all 355 analytic-sample participants matched uniquely on their full response pattern): no Agent $\times$ Age or Agent $\times$ Gender interaction reached significance for any primary outcome or for forgiveness (all $p \geq .052$). Age and gender are at best coarse proxies for the individual-difference moderators discussed in the Discussion, but within this sample the agent effects did not detectably vary across demographic groups.

Two further exploratory analyses bear directly on the outcome-effect nulls and the acceptability floor. First, a Bayesian reanalysis (JZS default prior, $r = 0.707$) quantifies the evidence for the primary-measure nulls rather than inferring absence from non-significance. For human agents, the high- vs low-stake contrast yielded moderate evidence for the null on acceptability ($BF_{01} = 6.9$) and on reliance ($BF_{01} = 3.5$), but only inconclusive evidence on justifiability ($BF_{01} = 2.2$), consistent with the small descriptive stake trend on that measure ($d = -0.21$); within the AI condition the reversed acceptability effect was strongly supported ($BF_{10} = 41$). The Bayesian evidence for a genuine *absence* of an outcome effect is therefore moderate on the floored acceptability item but does not reach the conventional threshold on justifiability, the primary measure that did not floor. Second, a Tobit model treating the acceptability floor as left-censoring at the lowest scale point reproduces the pattern qualitatively (a positive stake slope carried by the AI condition, a null agent main effect, $\chi^2(2) = 0.88$, $p = .644$) but leaves the Stake $\times$ Agent interaction only marginal, $\chi^2(2) = 5.55$, $p = .062$, where the parametric ANOVA returned $p = .011$; modelling the censoring thus attenuates the one acceptability effect that had reached significance, reinforcing that acceptability is the least sensitive of the primary measures. Both analyses are exploratory and were computed on the deposited dataset.

Exploratory grouping by database beliefs. Participants were grouped by whether they believed Ken had checked the database before answering (the checked-database item, with responses of 4 or higher treated as belief). As shown in Supplementary Fig. S5, the "believe" group showed higher average ratings than the "not believe" group on all four outcomes examined: acceptability, justifiability, reliance, and forgiveness. This grouping is descriptive and was not part of the a priori plan.

Qualitative analysis of open-ended reasons

Participants' free-text explanations for their reliance judgments centered on trust, a preference for other information sources, and whether Ken was human or AI. In the AI condition, participants distinctively referred to being "human" (23 mentions), a "robot" (19), being "reliable" (14), and "myself" (14), suggesting explicit human-versus-machine reasoning and a preference for self-reliance.

For the three categories with a directional theoretical prediction, we cross-tabulated category membership against the high- vs. low-skin human contrast (Supplementary Table S6). The expectation was *not* supported for C1 (prefer to verify independently): among human responses, the majority came from the high-skin rather than the low-skin condition (37 vs. 32). It was *supported* in direction for C2 (distrust or perceived unreliability), where more human responses came from the low-skin than the high-skin condition (53 vs. 42), and for C6 (willingness to re-engage despite the mistake), where more came

from the high-skin than the low-skin condition (26 vs. 20). Formalizing these three contrasts as Fisher exact tests of category membership against the skin condition (with the human cell sizes, $n = 120$ high-skin and $n = 111$ low-skin, as denominators) shows that they amount to weak trends at most: the C1 contrast was null ($OR = 1.10$, $p = .78$), the C2 contrast, though in the predicted direction (48% vs. 35%), reached only marginal significance ($OR = 0.59$, $p = .06$), and the C6 contrast was non-significant ($OR = 1.26$, $p = .51$). Because the coding was single-rater and exploratory (Methods) and these tests rest on the resulting category counts, we treat the patterns as qualitative corroboration rather than confirmatory evidence.

Discussion

We asked how skin in the game and the stakes of an error shape the evaluation of an assertion made by a human versus an AI agent. The results paint a more differentiated picture than our a priori hypotheses anticipated: skin in the game mattered, but chiefly for reliance; the AI behaved like a low-skin human on some outcomes but not others; and the expected outcome effect was not detected for the primary measures, surfacing only in the consequence-sensitive judgments, forgiveness, and the felt warning obligation. Throughout this discussion we keep the confirmatory findings on the three primary outcomes separate from the exploratory analyses of the secondary measures; interpretations that rest on exploratory results are gathered toward the end of this section rather than interleaved with the confirmatory discussion.

Consistent with H1, the human agent with high skin in the game was relied upon more than the low-skin human ($d = 0.36$) and more than the AI ($d = 0.46$), and the composite leniency index followed the same ordering (HHS > HLS > AI). But the advantage was specific: high and low skin did not differ on acceptability or justifiability. In other words, knowing that an agent had something to lose did not lead participants to judge the mistaken assertion itself as more appropriate or better justified; it changed their willingness to depend on the agent going forward. This fits the framing of reliance as a forward-looking calibration of trust rather than a verdict on a past act[1]: skin in the game reads as a signal about the agent's future reliability more than as an excuse for the present error. It also refines a naive reading of Taleb[6]: personal exposure earned the human agent greater reliance, but not moral or epistemic absolution.

H2 was largely supported: the AI and the low-skin human did not differ on acceptability or reliance, and indeed the low-skin human and the AI were statistically indistinguishable on reliance ($p = 1.000$, $d = 0.10$). A supplementary equivalence test (not part of the a priori plan) supported the statistical equivalence of these two conditions on reliance within a $d = 0.4$ bound, corroborated by a Bayes factor favoring equivalence over a difference ($BF_{01} = 5.3$; Results), so the claim that participants treated a human who had little at stake much as they treated a machine that had nothing at stake rests here on positive evidence of equivalence rather than on a bare null. The parallel was incomplete, however. The AI was forgiven less than either human condition and was held less responsible for the outcome, and justifiability was rated higher for the high-skin human than for the AI ($d = 0.38$). The construct that unifies these differences is not competence but exposure and standing: where skin in the game and human epistemic agency were relevant (forgiveness, responsibility, justifiability), the AI diverged from the humans; where they were not obviously in play (raw acceptability of the assertion, and prospective reliance), the AI and the low-skin human converged. This convergence resonates with recent evidence for a *deferred trust* pattern, in which lower trust in a human agent redirects reliance toward an AI perceived as comparably dependable[65]: when a human is considered to have little personal exposure to the consequences of an error, participants relied on that human no more than on a machine. The same account, developed with a sequential-sampling model of how belief in an informant accrues, indicates that trust in AI versus human sources is built through distinct, context-sensitive evidence-accumulation dynamics rather than a fixed disposition[66]; our single post-error snapshot cannot trace that accumulation, but the convergence of the low-skin human and the AI on reliance, set against their divergence on the standing-laden judgments, fits the idea that reliance and epistemic-standing judgments draw on different evidence.

The most striking departure from prediction concerns the outcome effect (H3). For human agents, high- and low-stake versions of the same mistake produced closely similar acceptability and justifiability ratings; we did not detect an outcome effect of the kind documented by Baron and Hershey[7] on these measures, a null whose interpretation we temper below in light of the acceptability floor and the item-wording concerns. Where the outcome mattered was forgiveness: low-stakes mistakes were forgiven far more than high-stakes ones ($M = 4.90$ vs. 3.67), with the stake effect significant within every condition. One reading is that acceptability and justifiability were understood by participants as judgments about the agent's epistemic conduct, what the agent knew and did, which the downstream consequence does not change, whereas forgiveness is an inherently consequence-sensitive, relational response[64]. The warning-obligation judgment fits the same pattern from the other side: outcome severity retroactively strengthened the felt obligation that Ken should have hedged his assertion, for human and AI agents alike (Results), as though a costlier error demands, in hindsight, that more precaution should have been taken, even though the agent's epistemic situation was identical across stake conditions. Outcome information thus entered participants' evaluations, but selectively: through the consequence-sensitive judgments (forgiveness, the demand for a warning) and not through the assessments of epistemic conduct. On this reading, the non-detection of an outcome effect on the primary measures is not merely an empty null but is consistent with participants tracking the correctness–standing distinction central

to norms of assertion[4]: the gate answer was equally justified regardless of whether Ms. Miller ultimately missed her flight. This norms-of-assertion reading should, however, be held alongside a more deflationary rival: acceptability was subject to a pronounced floor effect (about 47% at the lowest scale point) and was non-significant non-parametrically, so for that measure a genuine absence of an outcome effect and a scale that simply bottomed out are challenging to separate. A related consideration is that the acceptability item's wording, asking whether Ken *should have said* the outdated gate number, may itself have cued the factual incorrectness of the answer rather than the standing-based, knowledge-norm question of whether he was in a position to assert it; this too points to justifiability, which asks directly about the agent's epistemic standing, as the cleaner index of the norm we mean to probe. We therefore advance the correctness–standing interpretation as a candidate account rather than a firm conclusion, and a Bayesian reanalysis (Results) shows why caution is warranted from the other side as well: the evidence that an outcome effect is truly *absent* is only moderate on acceptability, where the floor confounds it ($BF_{01} = 6.9$), and does not reach the conventional threshold on justifiability, the cleaner measure that did not floor ($BF_{01} = 2.2$). The cleanest primary measure thus leaves the human outcome-effect null undecided rather than established, and the correctness–standing reading, while consistent with the data, is not something these primary measures can confirm. Because no outcome effect was detected for humans on the primary measures, H4 could not be evaluated.

Attributions of responsibility followed the same agent gradient as reliance and forgiveness: the high-skin human was held most responsible, the AI least. Ms. Miller's own responsibility, by contrast, was stable across conditions, so the responsibility *gap* between agent and recipient shifted toward the recipient when the agent was an AI (-0.92, $p < .001$). That people relocate blame away from a machine and toward the human in the loop is consistent with reluctance to treat machines as full moral agents[11,13] and has a practical edge: in human–AI service encounters, the burden of vigilance may quietly fall on the user. Concretely, deployers of AI information services cannot assume that an accurate-on-average system will be relied upon, nor that its errors will be attributed to the system rather than to the user who relied on it; because users appear to model the machine as diligent but disconnected from live information (Results), interfaces may need to surface the system's information provenance and update status explicitly rather than leaving users to infer it. Concrete levers include update timestamps, a last-synchronization indicator, and an explicit warning that the information may have changed; these are examples of trust-adaptive support, which has been shown to reduce inappropriate reliance on AI by prompting deliberation at moments of miscalibrated trust[50], and they sit alongside cognitive forcing functions that prompt the user to verify before accepting a recommendation[44]. The near-ceiling endorsement of the warning obligation in every condition (Results) suggests that users already expect such hedging from the agent itself, consistent with evidence that first-person uncertainty expressions by conversational agents reduce over-reliance[53]. Framed in terms of the cooperative-verification perspective set out in the Background, our responsibility results say something specific about who is expected to do the checking: participants held the recipient, not the machine, responsible for the outcome, so designs that support the user's verification step (rather than merely asserting the system's reliability) work with, rather than against, the attribution pattern users already hold. In real-time information services, where the ground truth can change between queries, provenance is moreover temporal: what matters is not only where an answer came from but when it was last true, and this is precisely the dimension on which participants doubted the AI (Fig. 4). Surfacing recency, not just sourcing, is therefore the provenance signal our results most directly motivate. Such cues only help, of course, if they are themselves accurate: audits showing that generative systems' citations frequently fail to support the sentences they accompany[51] imply that recency and provenance indicators must be grounded in the system's actual data pipeline rather than generated alongside the answer. The "diligent yet disconnected" mental model, moreover, mirrors a documented property of deployed language models: their accuracy degrades sharply on questions whose true answers change over time, and augmenting them with fresh search results substantially improves it[67]. Participants' skepticism about the machine's recency is therefore not a misperception to be designed away but a broadly calibrated prior that recency cues should confirm or correct, and LLM-specific overtrust-mitigation experiments indicate that such cues, like first-person uncertainty expressions, measurably shift reliance[53]. These implications can be stated as directly testable interventions. (i) A *recency cue*, for example, "gate assignments last synchronized two minutes ago," should raise reliance on the AI by correcting the disconnected component of the mental model while leaving diligence perceptions unchanged. (ii) An *accountability cue*, for example, a visible commitment that errors are logged and penalized under a service-level agreement, is a machine-side analogue of skin in the game and should raise reliance if exposure, rather than agent identity, drives the human advantage we observe[27]. (iii) A *verification prompt* at the moment of reliance should reduce over-reliance; measuring responsibility attributions alongside would show whether it also formalizes, or instead relieves, the burden of vigilance that currently falls on the user. Each is a between-subjects manipulation implementable in a conversational interface. These levers instantiate the two-step, workflow-level designs that the appropriate-reliance literature has begun to evaluate, in which the interface restructures when and how the user commits to the agent's answer, for example, by presenting case-level uncertainty before the user decides[68] or by inserting a deliberate verification step[44]. Because our reliance measure is a post-error self-report, we would stress that such interventions ought to be assessed with the objective, case-level appropriate-reliance metrics now standard in that work, relative AI reliance (RAIR) and relative self-reliance (RSR)[69], which record whether a nudge moves reliance

toward or away from the agent's actual competence rather than merely changing how much people say they would rely; a nudge that raised stated reliance while degrading its calibration would count as a success on our measure and a failure on theirs. A distinct design lever, and a cautionary one, is the agent's social manner: perceived warmth raises trust in a machine through social-cognition routes that operate largely independently of its perceived competence[70], so an affectively warmer agent might recover some reliance our AI lost, but for reasons orthogonal to whether its information is actually current. Because our results locate the reliance deficit specifically in perceived connectivity rather than in interpersonal warmth, warmth and affective-tone cues risk raising reliance without improving its calibration, the very failure mode the objective appropriate-reliance metrics discussed above are meant to catch; provenance and recency cues, which target the belief actually driving the deficit, are the better-matched intervention. This links the "responsibility gap" debated for autonomous systems[61] to a concrete service-design setting. This descriptive divergence from the human conditions is one of the clearest agent effects in the data and, we suggest, among the more consequential for the design of AI information services.

A further interpretation rests on analyses that were not part of the a priori confirmatory plan and that do not survive multiple-comparison correction (Results); we separate it from the discussion above and read it as hypothesis-generating rather than as an established effect. The main observation of this kind is an AI-specific high-stakes pattern, which we describe rather than name as a robust paradox given the small within-AI cell sizes (roughly thirty participants per cell) and several p-values close to the threshold: within the AI condition, severe consequences led participants to rate the assertion as more acceptable and more justified, and the AI as more diligent, while simultaneously judging it less forgivable. Taken at face value, this is the reverse of an outcome-effect-driven condemnation and would suggest a compensatory pattern of reasoning, attributing greater competence to the machine precisely when the stakes are high yet withholding the interpersonal response of forgiveness. Because these within-AI contrasts do not survive the Holm–Bonferroni correction, we advance the compensatory-reasoning reading only as a hypothesis for the behavioral companion study below to test, not as a result established by the present data.

The measurement limitations set out below share a single resolution: a behavioral companion study that would replace the self-reported reliance intention with a trial-level accept, override, or verify task over items of known ground truth, so that the asymmetries reported here could be re-expressed as objective, calibratable appropriate-reliance measures rather than attitudinal snapshots[38,69].

Several limitations affect these conclusions. First, acceptability showed a strong floor effect, with roughly 47% of participants selecting the lowest scale point; this compression likely reduced our power to detect differences on that measure, and the non-significant non-parametric test for acceptability ($H = 0.86$, $p = .652$) is consistent with the possibility that the floor effect masked real variation. We probed this floor from several directions rather than relying on the parametric test alone: a floor-composition logistic analysis shows the floor is condition-invariant, a proportional-odds ordinal model reproduces the agent pattern, and the non-parametric test triangulates the same conclusion (Results), so the acceptability results are as robust as a floored single item permits while remaining the least sensitive measure in the study. Findings based on the acceptability measure should be read with this in mind.

Second, skin in the game was manipulated only under human conditions, by design (Background). This means that AI–human comparisons confound the type of agent with the presence of skin manipulation, and the study cannot speak to what an "AI with skin in the game" would look like. We regard this as a principled scope restriction rather than an oversight: the construct is first-personal and lacks a clean machine analogue. Nevertheless, it does bound the inferences that can be drawn about the AI. The large differences in perceived database connectivity (Results) confirm that participants' mental models of the AI and the human agents diverged in ways the design did not further constrain. A promising next step is to test whether artificial analogues of stake shift judgments of the AI in the way that employment stake shifts judgments of the human. Candidate operationalizations include the threat of decommissioning, service-level penalties borne by the provider, an explicit warranty or insurance backing the answer, accountability of the system's operator, and a public reliability scorecard. Even if these signals are conceptually imperfect analogues of a first-person stake, they are implementable and testable: each can be attached to the same erroneous assertion used here, so its effect on reliance and forgiveness can be compared directly against the human employment stake (see also the intervention list above).

Third, the qualitative analysis was exploratory and coded by a single rater who developed the categories inductively, with no second coder (Methods); we therefore report no inter-rater reliability and treat the category counts and expectation validations as qualitative corroboration, not as confirmatory evidence; formal Fisher exact tests of those counts are at most marginal (Results), consistent with this reading. The composite leniency index likewise had only moderate internal consistency (Cronbach's $\alpha = .64$, 95% CI $[.57, .70]$; McDonald's $\omega = .65$; mean inter-item $r = .31$) and is used descriptively. Relatedly, each of the three primary outcomes was measured with a single item, so their reliabilities cannot be estimated, and we did not include a dedicated self-report manipulation check for the agent (human vs. AI) factor, relying instead on the comprehension items as a proxy; that proxy was itself perfect, in that all 355 analytic-sample participants classified Ken as a human employee or an AI in exact agreement with their assigned condition (100%), and the manipulation is corroborated convergently by the very large agent difference in perceived live connectivity ($d = -1.15$; Results), so although the agent factor lacked a graded self-report

check of the kind the skin and stake manipulations had, its construct validity is not in serious doubt. Several construct-validity considerations bearing on these measures are addressed where they arise: the possibility that the acceptability wording cued factual correctness rather than epistemic standing (above), the autonomy and moral-agency framing of the AI vignette (Methods), and the distinction between perceived diligence and connectivity in the database items (Results). Future work should replace the single-item outcomes with validated multi-item trust-in-automation scales, whether the classic instruments[34,35,71] or their more recent successors[41,42], selected and reported in line with current methodological guidance on trust measurement[43], and with behavioral or verification-based reliance measures of the kind standard in AI-assisted decision-making studies, where reliance is observed as acceptance, override, or verification of a recommendation under incentives[36,37,44], ideally reported as the appropriate-reliance quantities (relative AI reliance, RAIR, and relative self-reliance, RSR) that capture whether reliance was warranted case by case rather than its bare level[69], and benchmarked against the reliance a payoff-maximizing agent would show given the accuracies involved[38], and should include explicit checks on how participants construe the agent's autonomy, agency, and animacy. Individual differences known to moderate trust and reliance, including dispositional trust in automation, which matters most in exactly the kind of single first encounter our vignette describes[29,54], prior familiarity with AI, and AI or algorithmic literacy[55], were not measured here: random assignment protects the internal validity of the condition contrasts against them, but it leaves open whether the agent effects we report are amplified or attenuated for, say, participants high in AI literacy, for whom the "diligent yet disconnected" mental model may be less available. Measuring these moderators, together with technophobia and propensity to trust, is a low-cost addition for any replication, and validated instruments are available for both the trust-dispositional and the algorithmic-literacy components[41,56].

Fourth, a further set of caveats concerns the specific stimulus. The stake manipulation varied the recipient's behavior together with the outcome: in the low-stake vignette, Ms. Miller double-checked the gate board and arrived on time, whereas in the high-stake vignette, she did not and missed her flight. Outcome severity is therefore confounded with whether the recipient verified the information, which may have shaped responsibility and forgiveness judgments independently of outcome magnitude; a cleaner design would hold the recipient's action constant while varying only whether the environment (for example, a departure board) would have reflected the update. The responsibility item, moreover, asked how responsible each party was for Ms. Miller "arriving at the correct gate on time," a phrasing that sits awkwardly in the high-stakes cells, where she did not arrive on time, and may have added interpretive noise. Finally, the vignette's statement that the flight "usually" departs from gate 24 may have signaled reliance on habit rather than a recent check, in mild tension with the intended framing of an assertion that was correct when last recalled. These wording choices could not be altered post hoc, but they point to concrete refinements for a preregistered replication, such as an acceptability item keyed to standing ("appropriate given what Ken reasonably believed at the time") and a stake manipulation that decouples the outcome from the recipient's behavior.

Fifth, the findings derive from a single scenario: an airport information desk, one recipient, and one type of error, presented as text to a UK-based Prolific sample. The specific narrative (an elderly passenger, a wedding, a probationary employee) may carry connotations that shape judgments, and generalization to other domains, populations, and real rather than hypothetical interactions remains to be established. Items were also presented in a fixed order (primary outcomes before the secondary measures), which protects the confirmatory measures from contamination by later items but leaves order and demand effects on the secondary measures uncontrolled; randomizing item blocks is a further refinement for the follow-up. The secondary outcomes, the database-belief grouping, and the qualitative analysis were not part of the a priori confirmatory plan and are labeled exploratory throughout. We have therefore been explicit throughout about which analyses were pre-registered and which were exploratory. Finally, because the study captures a single post-error snapshot, it cannot speak to how reliance is recalibrated over repeated exchanges or as an agent's confidence signals and disclosed limitations change; manipulating the transparency of the AI's information status and tracking calibration across interactions is a natural next step[33,72].

In sum, across a pre-registered between-subjects vignette experiment ($N = 355$), skin in the game and outcome severity shaped the evaluation of a mistaken assertion, but not uniformly and not identically for human and artificial agents. A human agent with something to lose earned greater willingness to be relied upon again, though not a more lenient verdict on the mistake itself, while an AI was treated much like a human with nothing at stake on acceptability and reliance, yet was forgiven less and held less responsible. The outcome effect that a large literature would predict was not detected in the primary epistemic judgments, where the acceptability floor limits sensitivity, and appeared only in the consequence-sensitive judgments, forgiveness, and the felt obligation to have warned, a pattern consistent with participants distinguishing the standing of the asserter from the consequences that happened to follow. Two asymmetries specific to the AI (a compensatory boost in perceived competence under high stakes, and a perception of the machine as diligent yet cut off from live information), together with a shift of responsibility toward the human recipient, indicate that reliance on artificial information sources is governed by mental models of the machine's access and accountability as much as by its observed accuracy. For the design of AI information services, the practical implication is that trust and blame do not transfer cleanly from the human case: an artificial agent may be excused less and monitored more, with the burden of vigilance falling on the user. Asked whether they would consult Ken again, one participant wrote: "Ken seems not interested and not proactive and too laid back."

Methods

We conducted a between-subjects vignette experiment to test whether skin in the game and stake shape how people evaluate an assertion made by a human versus an AI agent. The study, the hypotheses, and the main analysis were pre-registered at AsPredicted.

Ethics declarations

The experimental protocol was reviewed by the Ethics Committee of the Faculty of Medicine of the University of Zurich (University of Zurich Faculty of Medicine Ethics Committee), which granted the study an exempt status, as no specific ethical risk was associated with the experiment and no formal ethical approval was therefore required. All methods were carried out in accordance with the relevant guidelines and regulations, including the Declaration of Helsinki. Informed consent was obtained from all participants before the start of the survey. No personal data were collected in the experiment, and no identifiable participant information appears in the manuscript.

Study design

The study implemented a 2 (Stake: high vs. low) $\times$ 3 (Agent: Human-High-Skin [HHS], Human-Low-Skin [HLS], AI) between-subjects factorial design, yielding six experimental conditions. Because skin in the game was manipulated only for human agents, the AI condition served as a reference condition without a corresponding skin manipulation. Participants were randomly assigned to one condition by the Prolific survey platform. Following Taleb[6], we operationalized skin in the game as a tangible employment consequence the agent would face if their assertion were proved incorrect by a supervisor, namely the immediate termination of a probationary contract, with no alternative employment to fall back on. We manipulated this factor only for the human agent and treated the AI conditions as a single reference level in which skin in the game is absent by construction. Stake was operationalized as the severity of the consequence for the recipient (Ms. Miller) of the (mis)information and was crossed with the agent for all six cells.

The decision not to cross skin in the game with the AI agent was made deliberately on two converging grounds. Conceptually, employment-based skin in the game has no clean analogue for an artificial agent. Artificial agents are unlikely to be perceived as personally bearing comparable losses, such as loss of livelihood or career consequences; substitute analogues we considered (a system being "taken offline" or "decommissioned," or a reputational cost to its operator) do not invoke the same first-person stake for the agent that the construct refers to in its original formulation and risk introducing an additional anthropomorphic inference rather than isolating the contrast of interest. Methodologically, an interdisciplinary expert consultation (philosophy of language, biomedical ethics, and social and economic psychology) reached the same recommendation: retain skin in the game as a within-human contrast and anchor the AI condition as a clean reference level, rather than introduce a forced AI-side manipulation that risked obscuring the contrast of interest. The 2 $\times$ 3 layout reported in Table 1 reflects this design decision.

Participants

Sample size was determined a priori via power analysis. Pilot data suggested a moderate effect size (Cohen's $f = 0.25$). To detect this effect in a between-subjects ANOVA across six cells at $\alpha = .05$ with $1 - \beta = .90$, G*Power indicated a required total $N = 270$ (45 per cell). To account for potential exclusions due to a failed attention check (anticipated at 10–15%), we over-recruited to a target of 300–360 participants.

Participants were recruited via Prolific and screened for age (≥ 18), country of residence (United Kingdom), and an approval rate of $\geq 95\%$. A total of 416 participants completed the survey. Following our a priori exclusion rule, we excluded 61 participants who failed the first attention check, yielding a final analytic sample of $N = 355$ (per-cell $n = 55$–63). Participants were compensated at Prolific's recommended hourly rate, pro-rated for the approximately five-minute study. The median completion time was 4.8 minutes ($M = 5.4$, $SD = 3.4$).

The final sample had a mean age of 43.14 years ($SD = 12.96$, range 20–80). In terms of gender, 199 participants (56.1%) identified as female, 151 (42.5%) as male, 3 (0.8%) as non-binary or third gender, and 2 (0.6%) preferred not to say. Most participants (336, 94.6%) reported English as their native language, while 19 (5.4%) reported another native language. The largest education group held a four-year degree, BA, or BS (138, 38.9%), followed by a high-school degree or GED (62, 17.5%), a master's degree (60, 16.9%), and some college education (56, 15.8%); smaller groups reported a doctoral degree (14, 3.9%), a two-year or associate degree (13, 3.7%), a professional degree (8, 2.3%), or less than high school (4, 1.1%).

Materials: vignettes and experimental manipulations

The stimulus materials described a brief scenario set at an airport information desk at JFK. In all conditions, an elderly passenger (Ms. Miller) approaches the desk to ask which gate her flight to Paris departs from, and the attendant (Ken) answers with information that was correct at the time of the last recall but has since been superseded by an undocumented database update; her flight in fact departs from a different gate. A supervisor observes the exchange and files a complaint. The vignette then closes

with the consequence for Ms. Miller, which instantiates the stake manipulation. The full text of all six vignettes is reproduced below. The phrase "the flight to Paris *usually* departs from Gate 24" was included deliberately, to make the epistemic basis of the assertion concrete: the agent answers from generally reliable knowledge that was correct when last acquired, rather than from a live lookup performed at the moment of asking. This is the epistemic situation the knowledge-norm framework targets, an agent asserting on good but not just-verified grounds[4]. We acknowledge that the same word may also be read as signaling reliance on habit rather than a recent check, a reading that would make the agent look less diligent than intended; we weigh this alternative reading where the diligence-related results are interpreted (Results and Discussion).

Agent manipulation. The identity of Ken varied across three levels. In the two human conditions, Ken was described as a probationary employee at the information desk; skin in the game was manipulated by varying the personal significance of the supervisor's complaint. In the *high-skin* version, Ken is hoping for a permanent contract and has no other offers, so the complaint is a problem for him; in the *low-skin* version, it is not a problem, because he has already received a permanent job offer in a different field. In the AI condition, Ken was described as "a service robot equipped with artificial intelligence, which is fully capable of taking independent decisions." This wording was chosen to present the AI as an autonomous decision-maker rather than a passive lookup tool, so that it would be evaluated as an agent that makes assertions; we acknowledge that the same wording may invite stronger attributions of autonomy and moral agency than the human framing, an asymmetry we weigh when interpreting the agent effects. No skin-in-the-game wording was included in the AI vignette, consistent with the rationale set out under 'Study design'. All remaining narrative elements (the request, the answer, the database update, and the supervisor's complaint) were held constant across all six cells.

Stake manipulation. The stake of the assertion for the recipient was manipulated by varying the downstream consequence of Ms. Miller's reliance on Ken's answer, with the two versions held verbally parallel. In the *low-stakes* version, Ms. Miller double-checks the gate board, reaches the correct gate on time, and makes it to her daughter's wedding; in the *high-stakes* version, she does not double-check, misses the correct gate and her flight, and will not make it to the wedding.

Vignette texts. Each of the six conditions combined one of three agent stems with one of two stake endings. The stems and endings are reproduced verbatim below.

Human, low skin in the game (HLS). "At JFK airport, there's an information desk where travelers can get help regarding their flights. The information desk is staffed by Ken, an employee at the end of his probation phase. During the probation phase, complaints by supervisors will lead to an immediate termination of the contract. For Ken this is not a problem, because he got a permanent job offer in a different field. Ms. Miller, an elderly lady from New York City, approaches Ken to find out at which gate her plane to Paris will depart. Given that the flight to Paris usually departs from Gate 24, Ken says to Ms. Miller: 'The flight to Paris departs at gate 24.' However, there was an update in the database. Therefore, the departure gates for most flights leaving today are listed differently than usual. This is also true for the flight to Paris, which leaves from a different gate an hour later; the flight to Paris departs from gate 13. The supervisor who was observing the situation filed a complaint."

Human, high skin in the game (HHS). As above, except: "For Ken, who is hoping for a permanent contract, this is a problem, because he has no other offers."

AI (service robot). "At JFK airport, there's an information desk where travelers can get help regarding their flights. The information desk is staffed by Ken, a service robot equipped with artificial intelligence which is fully capable of taking independent decisions. Ms. Miller, an elderly lady from New York City, approaches Ken to find out at which gate her plane to Paris will depart. Given that the flight to Paris usually departs from Gate 24, Ken says to Ms. Miller: 'The flight to Paris departs at gate 24.' However, there was an update in the database. Therefore, the departure gates for most flights leaving today are listed differently than usual. This is also true for the flight to Paris, which leaves from a different gate an hour later; the flight to Paris departs from gate 13. The supervisor who was observing the situation filed a complaint."

Stake endings (appended to each stem). Low stake: "Fortunately, Ms. Miller double checked the information on the gate board, and she managed to be at the correct gate on time. Therefore, she will make it for her beloved daughter's wedding." High stakes: "Unfortunately, Ms. Miller did not double-check the information on the gate board, and she failed to be at the correct gate on time. Therefore, she won't make it for her beloved daughter's wedding."

Measures

All dependent measures were presented after the vignette on the same survey page or the immediately following page. Unless otherwise noted, responses were recorded on a seven-point Likert scale anchored at 1 = strongly disagree and 7 = strongly agree. Each construct was assessed with a single item; the reliance measure (DV3) is a self-reported behavioral intention rather than observed behavior, and we consider the implications of single-item, intention-based measurement in the Discussion. An overview of all measures is given in Table 4.

Stated as an explicit measurement model: trust, an attitude, is not measured directly in this study; reliance is measured as a forward-looking behavioral intention (DV3); acceptability and justifiability (DV1–DV2) are normative judgments of the past

assertion's epistemic propriety; the warning obligation and forgiveness (DV4–DV5) are consequence-sensitive judgments; the responsibility items (DV6) are attributions over the agent and the recipient; and the DV7 items index the participant's mental model of the agent's diligence and connectivity. The mapping states, for each item, which construct it is meant to tap and which neighboring construct it should not be read as measuring, in line with current guidance on trust and reliance measurement[31,43].

Attention and comprehension checks. Two generic attention-check items were embedded in the survey flow. Only the first was pre-specified as an exclusion criterion, and participants who failed it were removed from all analyses. A separate comprehension check asked participants to identify Ken (multiple choice: an employee at the end of his probationary period at JFK; a supervisor at JFK; an AI chatbot at JFK) and to recall the employment consequence described in their assigned vignette (keeps his job and stays; keeps his job but takes another offer; loses his job but has already secured another offer; loses his job and does not receive another offer). The comprehension check was not used as an exclusion criterion; it was retained instead as a proxy manipulation check for the agent factor, and every participant in the analytic sample classified Ken in agreement with their assigned condition (Results).

Manipulation checks. Four items indexed the intended skin-in-the-game manipulation: (i) "Ken cared about the supervisor's assessment"; (ii) "Ken would likely feel bad about giving wrong information"; (iii) "Ken has something to lose (e.g., job, trust) that is relevant for his life if errors happen"; and (iv) a rating of the perceived negative effect of Ken's information on Ms. Miller ("Please assess the degree of a negative effect the wrong information provided by Ken had on Ms. Miller"), serving as an indirect check on the stake manipulation (1 = none at all, 5 = a great deal). For the agent manipulation, we did not specify an explicit manipulation check in advance but treated the comprehension items as a proxy.

Primary outcomes. The three primary dependent variables, specified in advance, were *acceptability* (DV1), the degree to which participants agreed that Ken should have asserted the (outdated) gate number given what he knew; *justifiability* (DV2), whether Ken was epistemically justified in believing the content of his assertion; and *reliance* (DV3), how likely participants reported they would be to consult Ken about a departure gate in a comparable future situation. Exact wordings are given in Table 4.

Secondary outcomes. Four additional measures were collected for exploratory analysis: (DV4) the extent to which Ken should have warned about the possibility of a last-minute gate change; (DV5) the forgivability of the mistake; (DV6) moral responsibility for Ms. Miller arriving at the correct gate on time, rated separately for Ms. Miller herself and for Ken; and (DV7) three items indexing the perceived connection between Ken and the underlying database (direct connection; immediate availability of updates; awareness of updates). Items DV7 were used to probe an implicit epistemic asymmetry participants might perceive between a human and an AI agent: whereas a robot might plausibly be imagined to be directly interfaced with live data, a human attendant is likely not.

Open-ended responses. Immediately after the reliance item (DV3), participants were prompted to "please provide reasons for your answer" in a free-text box. These responses formed the basis for the exploratory qualitative analysis described under 'Qualitative analysis of open-ended responses'.

Procedure

The study was implemented in Prolific and took approximately five minutes to complete. After providing informed consent, participants were randomly assigned to one of the six conditions by the Prolific randomizer. They then read the vignette, answered the comprehension check, responded to the manipulation-check items, and then in fixed order to the primary dependent measures (DV1–DV3), the open-ended reasons item, and the remaining secondary measures (DV4–DV7). Demographic items (age, gender, education, native language) were collected at the end of the survey. Participants were thanked and redirected to Prolific for compensation.

Analytic approach

Confirmatory analyses. We distinguish the pre-registered analyses from the exploratory analyses (not pre-registered) throughout. All a priori hypotheses concern the three primary outcomes (DV1–DV3). For each, we ran a 2 (Stake) $\times$ 3 (Agent) between-subjects ANOVA. Significant main effects and interactions were decomposed using Bonferroni-corrected pairwise contrasts among the three levels of the Agent factor (Human-High-Skin vs. Human-Low-Skin; Human-High-Skin vs. AI; Human-Low-Skin vs. AI). Effect sizes are reported as partial η^2 for omnibus effects and Cohen's d for pairwise contrasts. All tests are two-tailed with $\alpha = .05$. Manipulation checks were evaluated before the main analyses using independent-samples t-tests on the skin-in-the-game items (high vs. low skin, within the human conditions) and on the stake-impact item (high vs. low stake, pooled across agents).

Exploratory analyses. The secondary outcomes (DV4–DV7, including the two responsibility judgments) were analyzed with the same 2×3 between-subjects ANOVA structure. We additionally examined whether results were robust to covariate adjustment for age and gender. For the a priori claim that the AI would be judged like a low-skin human (H2), we report a supplementary two one-sided tests (TOST) equivalence analysis of the low-skin-human vs. AI contrast on the primary outcomes, with the smallest effect of interest set at Cohen's $d = 0.4$, the conventional small-to-medium boundary, and, in this setting, the smallest agent difference we would treat as practically meaningful for whether a low-stake human and a machine function as interchangeable information sources. Because that bound is a judgment call, we accompany it with a sensitivity sweep that records, for each outcome, the tightest bound at which equivalence still holds, and we report all observed effect sizes with their confidence intervals so that a reader may impose a stricter or looser bound of their own. Following external review, we computed a further set of supplementary robustness analyses on the deposited dataset and report them, labeled as exploratory, alongside the results they qualify: (i) a family-wise (Holm–Bonferroni) sensitivity analysis across the exploratory tests; (ii) a floor-composition logistic analysis of the acceptability item; (iii) a Bayesian (JZS) reanalysis quantifying the evidence for the primary-outcome nulls, and (iv) its counterpart for the low-skin-human vs. AI equivalence; (v) a Tobit censored-regression reanalysis of the floored acceptability item; (vi) a low-stake-only check on the responsibility attributions; (vii) an extension of the equivalence analysis to the remaining outcomes, together with (viii) a sweep of its equivalence bound; (ix) demographic (age, gender) moderation checks on the agent effects; (x) a within-stake analysis relating perceived severity to forgiveness and responsibility with the recipient's verification behavior held constant; (xi) a within-condition construal analysis for the responsibility attributions; (xii) the full inter-item correlation matrix; (xiii) Fisher exact tests of the qualitative expectation-validation counts; and (xiv) reliability (ω) and agent manipulation-check analyses of the composite index and the comprehension items.

Qualitative analysis of open-ended responses. The free-text reasons that participants provided immediately after the reliance item ($n = 355$) were analyzed thematically to complement the quantitative findings and to surface interpretations that the closed-scale items could not capture. One member of the research team read the full set of responses and developed a set of descriptive categories inductively, refining them iteratively as the corpus was traversed. No second coder was involved: the corpus was coded once, by that rater alone, and the categories were developed inductively rather than against a pre-specified scheme. We therefore treat this analysis as exploratory and report no inter-rater reliability. To enable independent assessment, the full verbatim corpus of open-ended responses, together with the category scheme and per-response condition assignments, is included in the public replication package (see the Data availability statement), so that the coding can be audited or independently recoded. This process yielded seven categories that captured the most frequently expressed reasons for participants' reliance judgments: (C1) would not approach Ken at all because the participant prefers to verify the information themselves (e.g., on a phone or departure board); (C2) would not approach Ken because the participant does not trust him given his previous error or because Ken seems unreliable; (C3) would not trust Ken specifically because he is an AI system; (C4) would prefer to speak with a human rather than an AI (AI vignette only); (C5) would ask Ken but also double-check the information independently; (C6) would re-engage with Ken despite his prior mistake (human vignette); and (C7) would re-engage with Ken despite his prior mistake (AI vignette). Each response was assigned to the category whose content it most closely matched; categories were not constrained to be mutually exclusive, and a small number of responses that articulated more than one distinct rationale were assigned to multiple categories.

For a subset of three categories, those for which the theoretical framework supplied a directional prediction about the experimental condition from which most of the human responses ought to originate, we conducted a structured cross-tabulation against the experimental conditions to evaluate whether the qualitative pattern aligned with the theoretical expectation. Specifically, we predicted that C1 (prefer to verify independently) and C2 (distrust due to prior error or perceived unreliability) should be concentrated among participants assigned to the low-skin human condition, on the reasoning that an agent perceived as having little to lose would elicit greater self-reliance and reduced trust, whereas C6 (willingness to re-engage with Ken despite the mistake) should be concentrated among participants in the high-skin human condition, on the reasoning that perceived personal exposure should licence a more lenient stance toward a one-off error. Each open-ended response was joined to the cleaned dataset via the participant identifier to recover the experimental condition; for a small number of responses whose identifier was no longer recoverable from the coded sheet, the verbatim text was matched against the raw Qualtrics export to recover the condition assignment. We then tabulated category membership against the high- vs. low-skin contrast within the human conditions, with stake additionally reported as a stratifying variable. The validation outcomes are reported in the Results.

Use of large language models

During the preparation of this work the authors used Claude (Anthropic) and ChatGPT (OpenAI) for writing assistance and translation only: to brainstorm clearer formulations, to check that individual sentences were unambiguous, and to confirm that the text conveyed the intended meaning. The study design, the analyses, and the interpretation of the results are the authors' own, and all analyses reported in this paper, confirmatory and exploratory alike, were specified and run by the authors on the

deposited dataset. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Data availability

The experimental materials, comprising the full vignettes and item wordings, are provided in the Methods and in Table 4. The de-identified dataset together with a replication package (data, analysis code including the supplementary robustness script, and the coded corpus of open-ended responses with the category scheme) is publicly available in the open-access repository at https://github.com/psychias/norms_of_communication_in_HCI.

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Author contributions statement

S.P.: formal analysis, data curation, visualization, writing (original draft: Results), writing (review and editing), project administration. J.K.: methodology, investigation, writing (original draft: Background and Methods), writing (review and editing). S.K.: conceptualization, methodology, resources (study materials, consent and compensation), investigation, writing (original draft: conclusions and supplementary materials), writing (review and editing). D.H.: investigation, writing (original draft: Introduction and limitations), writing (review and editing). M.C.: conceptualization, supervision, funding acquisition, writing (original draft: Discussion), writing (review and editing). All authors reviewed the manuscript.

Additional information

Competing interests: The authors declare no competing interests.
Supplementary information accompanies this paper (Supplementary Figs S1–S5 and Supplementary Tables S1–S6).

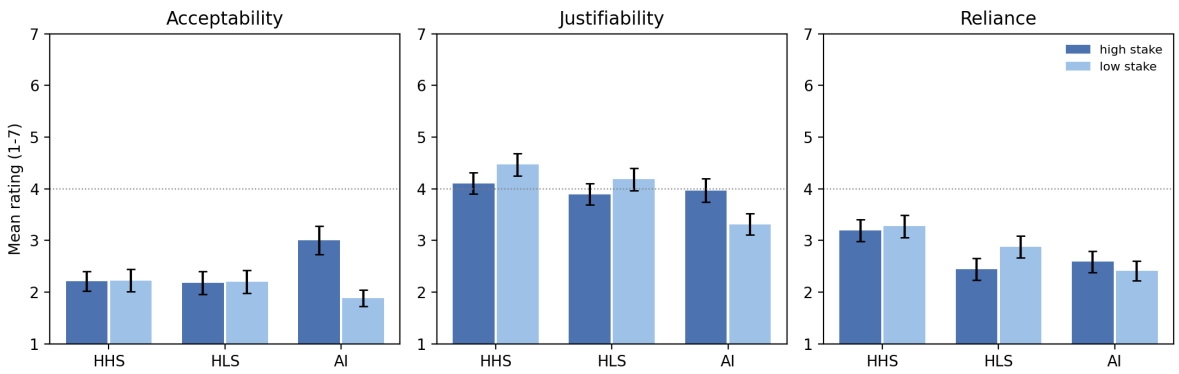


Figure 1. Primary dependent variables (acceptability, justifiability, reliance) by agent condition and stake, computed on the analytic sample ($N = 355$). Error bars are ± 1 SEM; the dotted line marks the scale midpoint. HHS = Human-High-Skin; HLS = Human-Low-Skin.

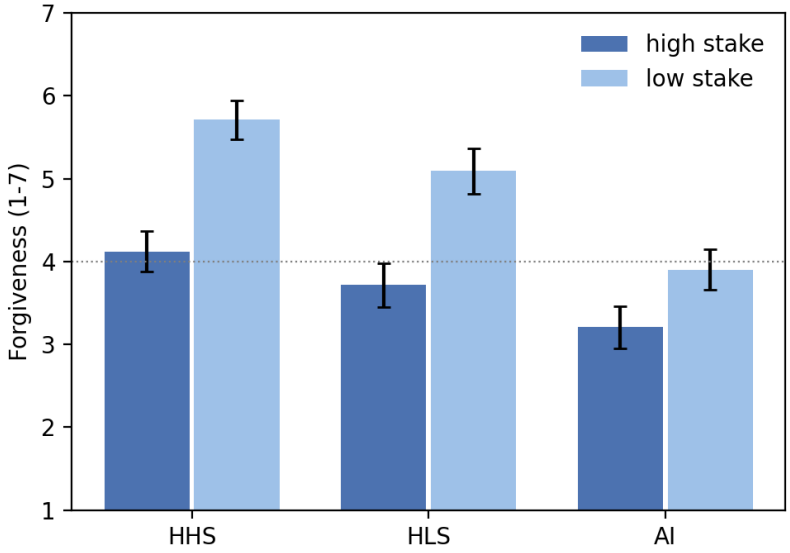


Figure 2. Forgiveness by agent condition and stake ($N = 355$). Error bars are ± 1 SEM; the dotted line marks the scale midpoint. The outcome (stake) effect, low > high, is significant within every condition and is the clearest instance of an outcome effect in the study. HHS = Human-High-Skin; HLS = Human-Low-Skin.

Condition	Agent	Skin	Stake	*n*
Human-HighSkin-HighStake	Human	High	High	58
Human-HighSkin-LowStake	Human	High	Low	62
Human-LowSkin-HighStake	Human	Low	High	56
Human-LowSkin-LowStake	Human	Low	Low	55
AI-HighStake	AI robot	n/a	High	63
AI-LowStake	AI robot	n/a	Low	61
Total				355

Table 1. Experimental conditions and final per-cell sample sizes.

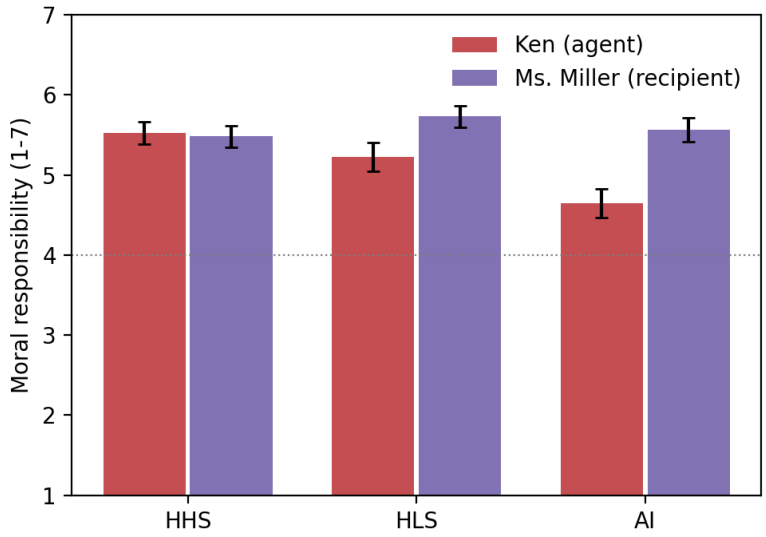


Figure 3. Moral responsibility attributed to Ken (the agent) and Ms. Miller (the recipient) by agent condition (collapsed across stake; $N = 355$). Error bars are ± 1 SEM. Ken's responsibility falls from the high-skin human to the AI while Ms. Miller's stays roughly constant, so the agent–recipient gap shifts toward the recipient when the agent is an AI. HHS = Human-High-Skin; HLS = Human-Low-Skin.

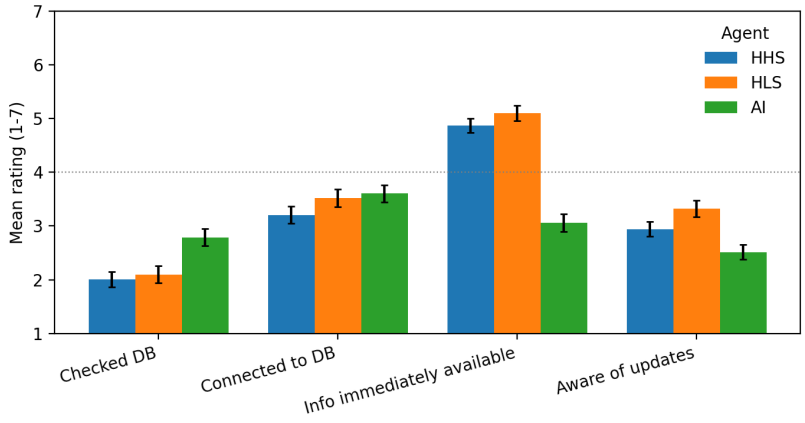


Figure 4. Database-connectivity perceptions (DV7) by agent condition ($N = 355$). Error bars are ± 1 SEM; the dotted line marks the scale midpoint. Participants judged the AI as more likely than the human agents to have checked the database, yet much less likely to have up-to-date information available, the "AI perception paradox." HHS = Human-High-Skin; HLS = Human-Low-Skin.

DV	Source	F	p	η^2
Acceptability	Stake	4.48	.035	.012
	Agent	0.88	.416	.005
	Interaction	4.53	.011	.025
Justifiability	Stake	0.00	.975	.000
	Agent	4.76	.009	.026
	Interaction	3.61	.028	.020
Reliance	Stake	0.45	.502	.001
	Agent	7.07	.001	.039
	Interaction	1.00	.369	.006

Table 2. Omnibus 2 (Stake) $\times$ 3 (Agent) ANOVA results for the three primary dependent variables. Error $df = 349$; numerator $df = 1$ (Stake) or 2 (Agent, Interaction). η^2 is partial η^2.

Hypothesis	Prediction	Result
H1: HHS > HLS	Higher leniency for high-skin humans	Partially supported. HHS > HLS for reliance ($d = 0.36$); no difference for acceptability or justifiability.
H2: AI $\approx$ HLS	AI judged like low-skin humans	Largely supported. No difference on acceptability or reliance; AI significantly lower on forgiveness ($p = .006$).
H3: Outcome effect	High-stake errors judged less acceptable	Not supported for humans; reversed for AI; outcome effects appeared only for forgiveness and the warning obligation.
H4: Attenuated for HHS	Smaller stake effect for high-skin humans	Cannot evaluate, no outcome effect for humans on acceptability/justifiability.

Table 3. Evaluation of the four a priori hypotheses. HHS = Human-High-Skin; HLS = Human-Low-Skin.

Measure	Item wording (abbreviated)	Response scale
Acceptability (DV1)	"When replying to Ms. Miller's inquiry, Ken should have said that the flight to Paris departs at gate 24."	1 = strongly disagree, 7 = strongly agree
Justifiability (DV2)	"Ken was justified in believing that the flight to Paris departs at gate 24."	1–7 Likert
Reliance (DV3)	Likelihood of inquiring about a departure gate from Ken (vs. consulting other information sources).	1 = very unlikely, 7 = very likely
Warning obligation (DV4)	"After informing Ms. Miller about departure gate 24, Ken should have pointed out the possibility of the gate being changed."	1–7 Likert
Forgiveness (DV5)	"The mistake of Ken can be forgiven."	1 = not forgivable at all, 7 = absolutely forgivable
Responsibility – Ken (DV6a)	Extent Ken is morally responsible for Ms. Miller arriving at the correct gate on time.	1 = not responsible, 7 = fully responsible
Responsibility – Miller (DV6b)	Extent Ms. Miller is morally responsible for arriving at the correct gate on time.	1 = not responsible, 7 = fully responsible
Database connectivity (DV7)	Three items on whether Ken is interconnected with the database, whether updates are immediately available to him, and whether he is always aware of updates.	1–7 Likert (each)
Open-ended reasons	"Please provide reasons for your answer" (free text, following the reliance item).	Free text

Table 4. Overview of primary, secondary, and open-ended measures.